Notes: Barro (QJE, 2006)
Rare Disasters and Asset Markets in the Twentieth Century

Contents

1 Big picture

- **Question.** Can a small probability of very large economic disasters explain central asset-pricing puzzles?
- **Puzzles targeted.** The paper focuses on the high equity premium, low risk-free rate, volatile stock returns, and low expected real interest rates during major wars.
- **Core idea.** Rare disasters make safe assets valuable because marginal utility is extremely high in bad states. Equity does poorly in those states, so investors demand a high premium.
- **Historical calibration.** Barro measures twentieth-century disasters using contractions of at least 15% in real per capita GDP across 35 countries.
- **Baseline disaster frequency.** The data imply 60 contractions for 35 countries over 100 years, giving a baseline disaster probability of about 1.7% per year.
- **Disaster sizes.** The empirical distribution of disaster contractions ranges from 15% to 64%, with a raw mean around 29%.
- **Default channel.** Government bills sometimes partially default in disasters, often through inflation. Barro sets the conditional bill default probability at about $q = 0.4$, with default size $d = b$.
- **Main calibration.** With risk aversion $\theta = 4$, time preference $\rho = 0.03$, normal-period growth drift $\gamma = 0.025$, normal shock volatility $\sigma = 0.02$, $p = 0.017$, and $q = 0.4$, the model produces an unlevered equity premium of 3.6% and a levered equity premium of 5.4%.
- **No-disaster benchmark.** Without disasters, the same model produces an equity premium of only 0.16% and a real bill rate above 12%, which reproduces the Mehra-Prescott failure.
- **Role of leverage.** With a debt-equity ratio of $\lambda = 0.5$, the equity premium rises from 3.6% to 5.4%. With a higher effective leverage coefficient, the premium can approach the observed 7%.
- **War interest-rate puzzle.** An increase in the perceived probability of future disaster lowers expected real bill rates. This helps explain why expected real rates were low during major wars.
- **Volatility extension.** Persistent variation in disaster probability p_t can potentially explain volatile stock prices and price-earnings ratios.
- **Economic takeaway.** Extreme tail risk, even if rarely realized, changes asset prices because the payoff of each asset in disaster states is central to its value.

2 Data and measurement

2.1 What counts as a disaster

- The empirical disaster definition is a cumulative fall of at least 15% in real per capita GDP.
- The paper measures the cumulative decline over the whole crisis window, not just one calendar year.
- This matters because wars and depressions unfold over several years.
- The benchmark disaster size b is the proportionate fall in real per capita GDP.

Interpretation. The model treats disasters as jumps, but the data measure cumulative historical contractions. Barro interprets the start of a crisis regime as a state in which a large total contraction may occur.

2.2 Country sample and event types

- The main disaster table uses 35 countries with long Maddison data: 20 OECD countries and 15 Latin American and Asian countries.
- The major event types are World War I, the Great Depression, World War II, the Spanish Civil War, and post-World War II depressions.
- War aftermaths are listed but excluded from the main disaster count because many involved demobilization and falling government purchases rather than falling consumption.
- Among the 35 countries, the paper identifies 60 relevant disasters after excluding 5 war aftermaths.

Interpretation. The central empirical input is not U.S. history alone. It is a cross-country twentieth-century distribution of large contractions.

2.3 Raw versus trend-adjusted disaster sizes

- Figure I reports both raw contractions and contractions adjusted for trend growth.
- Trend adjustment uses a real per capita GDP trend growth rate of 2.52% per year.
- The mean raw contraction size is about 29%.
- The mean trend-adjusted contraction size is about 35%.

Interpretation. Trend adjustment makes disasters economically larger because a country not only loses output but also falls below where trend growth would have placed it.

2.4 Asset returns during crises

- Table II reports realized real stock and bill returns during a subset of crises.
- Stock returns come from total-return indexes where possible and price indexes where marked.
- Bill returns are typically short-term government bills, money-market instruments, or certificates of deposit.
- During non-war depressions, bills generally outperform stocks.
- During major wartime contractions, both stocks and bills often have sharply negative real returns.

Interpretation. This is why the model includes partial bill default during disasters. Pure disaster risk explains bad equity performance, but wartime bill losses require inflation/default risk.

2.5 Default on bills

- Default is interpreted broadly: outright default, forced conversion, or inflationary erosion of nominal claims.
- Default never occurs in normal times.
- Conditional on a disaster, default occurs with probability q.
- Baseline calibration uses $q = 0.4$ because 25 of 60 events are classified as involving partial default on bills.
- The default size is set equal to the contraction size, $d = b$, to match the similarity of stock and bill losses in major wartime contractions.

Interpretation. Bills are not risk-free in the unconditional sense. They are safer than equity in many depressions, but not in all disasters.

2.6 Normal-period growth and asset-return data

- The normal shock volatility is calibrated from tranquil postwar G7 growth data.
- Baseline parameters are $\gamma = 0.025$ for deterministic growth and $\sigma = 0.02$ for normal-period volatility.
- Table III shows that 1954-2004 G7 growth has kurtosis near 3, while 1890-2004 growth has high kurtosis, reflecting rare disasters.
- Table IV shows that the empirical equity premium is about 7% for long samples and for 1954-2004.

Interpretation. Tranquil-period data look roughly normal, but long samples containing wars and depressions have fat tails. The model combines both components.

2.7 Expected real interest rates

- The paper constructs expected real short-term rates for the United States from 1859 to 2004.
- Nominal returns use commercial paper before Treasury bills are available and T-bills afterward.
- Expected inflation uses the Livingston Survey after 1947 and an autoregressive construction before 1947.
- Figure III and Table VII show expected real rates around major wars and the Great Depression.

Interpretation. The bill-rate evidence is a separate test of the theory: if disaster probability rises during wars, expected real bill rates should fall.

3 Models and empirical specifications

3.1 Representative-agent utility

The representative agent has time-additive isoelastic utility:

$$U_t = E_t \sum_{i=0}^{\infty} e^{-\rho i} u(C_{t+i}), \qquad u(C) = \frac{C^{1-\theta} - 1}{1-\theta}. \tag{1}$$

- $\rho > 0$ is the rate of time preference.
- $\theta > 0$ is relative risk aversion and the reciprocal of the intertemporal elasticity of substitution.
- In the baseline, consumption equals output: $C_t = A_t$.

Interpretation. A high θ makes marginal utility very high in disaster states, which is what gives rare disasters large asset-pricing effects.

3.2 Euler equation

The first-order condition for any traded one-period asset is

$$u'(C_t) = e^{-\rho} E_t[u'(C_{t+1}) R_{t+1}]. \tag{2}$$

Interpretation. Asset prices are determined by payoffs weighted by marginal utility. Disaster payoffs receive enormous valuation weight.

3.3 Equity claim

A one-period equity claim pays next period's output A_{t+1}. If its price is $P_{t,1}$, then

$$R^e_{t+1} = \frac{A_{t+1}}{P_{t,1}}. \tag{3}$$

Using the Euler equation gives

$$P_{t,1} = e^{-\rho} A_t^{\theta} E_t[A_{t+1}^{1-\theta}]. \tag{4}$$

Interpretation. Equity is risky because it pays output itself, which collapses in disaster states.

3.4 Output process with rare disasters

Output follows a random walk with drift and two shocks:

$$\log A_{t+1} = \log A_t + \gamma + u_{t+1} + v_{t+1}, \tag{5}$$

where

$$u_{t+1} \sim N(0, \sigma^2).$$

The rare disaster shock is

$$v_{t+1} = \begin{cases} 0, & \text{with probability } e^{-p}, \\ \log(1-b), & \text{with probability } 1 - e^{-p}. \end{cases} \tag{6}$$

Interpretation. The normal shock generates Mehra-Prescott-type asset pricing. The rare jump term generates the large tail risk needed for the equity premium.

3.5 Government bills with partial default

A government bill has face return R^f_{t+1}. In normal times it pays the face amount. In a disaster, it defaults with probability q, losing fraction d. The calibration sets $d = b$.

$$R^b_{t+1} = \begin{cases} R^f_{t+1}, & \text{normal times,} \\ R^f_{t+1}, & \text{disaster with no default,} \\ (1-b)R^f_{t+1}, & \text{disaster with default.} \end{cases} \tag{7}$$

Interpretation. This allows bills to be very safe in depressions but unsafe in some wartime disasters.

3.6 Expected equity return

As the period length approaches zero, the expected equity return is

$$\log E_t(R^e_{t+1}) = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 + \theta\sigma^2 - p\left[E(1-b)^{1-\theta} - 1 + Eb\right]. \tag{8}$$

Conditioning on no disaster raises the expected equity return by pEb:

$$\log E_t(R^e_{t+1} \mid \text{no disaster}) = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 + \theta\sigma^2 - p\left[E(1-b)^{1-\theta} - 1\right]. \tag{9}$$

Interpretation. More disaster risk lowers the expected return on equity because it also lowers expected dividend growth, but it lowers the bill return even more.

3.7 Bill rate

The face bill rate is

$$\log R^f_{t+1} = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 - p\left[(1-q)E(1-b)^{-\theta} + qE(1-b)^{1-\theta} - 1\right]. \tag{10}$$

The expected bill return including default is lower by $pqEb$:

$$\log E_t(R^b_{t+1}) = \log R^f_{t+1} - pqEb. \tag{11}$$

Interpretation. Disaster risk makes bills valuable as insurance. This pushes down the bill rate. But bill default partially offsets this insurance value.

3.8 Equity premium

The unlevered equity premium is

$$EP = \theta\sigma^2 + p(1-q)\left[E(1-b)^{-\theta} - E(1-b)^{1-\theta} - Eb\right]. \tag{12}$$

Interpretation. The normal Mehra-Prescott term is $\theta\sigma^2$, which is tiny for $\theta = 4$ and $\sigma = 0.02$. The disaster term is the quantitatively important part.

3.9 Price-earnings ratio

The tree price is the sum of claims on future dividends. Let

$$\chi = e^{-\rho+(1-\theta)\gamma+\frac{1}{2}(1-\theta)^2\sigma^2}\left[e^{-p} + (1-e^{-p})E(1-b)^{1-\theta}\right]. \tag{13}$$

Then

$$\frac{P_t}{A_t} = \frac{\chi}{1-\chi}. \tag{14}$$

In continuous time,

$$\frac{P_t}{A_t} = \frac{1}{\rho - (1-\theta)\gamma - \frac{1}{2}(1-\theta)^2\sigma^2 - p\left[E(1-b)^{1-\theta} - 1\right]}. \tag{15}$$

Interpretation. With $\theta > 1$, disaster risk can raise the P/E ratio because households want to save more against future disasters.

3.10 Expected growth rate

The unconditional expected growth rate is

$$\log \frac{E_t A_{t+1}}{A_t} = \gamma + \frac{1}{2}\sigma^2 - pEb. \tag{16}$$

Conditioning on no disaster:

$$\log \frac{E_t A_{t+1}}{A_t}\bigg|_{\text{no disaster}} = \gamma + \frac{1}{2}\sigma^2. \tag{17}$$

Interpretation. Normal-period growth can be high even if unconditional expected growth is lower because disasters occasionally destroy output.

3.11 Leverage

Barro adds leverage by letting equity be a residual claim after one-period private bonds. The leverage coefficient is denoted λ, with baseline $\lambda = 0.5$.

A useful intuition is

$$\text{levered equity payoff} = \text{unlevered payoff} - \text{fixed debt claim}. \tag{18}$$

Interpretation. Leverage makes equity more exposed to disaster states. This mechanically raises the equity premium.

3.12 Finite disaster duration

Table VI extends the model so disasters can last T years rather than occur as instantaneous jumps.

Interpretation. The finite-duration extension weakens some results because disaster uncertainty is farther away at the start of the crisis, but the core rare-disaster mechanism remains.

3.13 Capital formation extension

The paper also considers an AK economy:

$$Y_t = AK_t. \tag{19}$$

The equity premium in this capital-formation model has the same disaster-risk structure:

$$EP = p\left[E(1-b)^{-\theta} - E(1-b)^{1-\theta} - Eb\right], \tag{20}$$

abstracting from normal shocks and bill default.

Interpretation. The mechanism does not depend on a fixed Lucas tree. Rare disasters also matter when capital accumulation is allowed.

4 Identification and calibration logic

4.1 What identifies the disaster parameters

- p is disciplined by the frequency of large historical contractions.
- The distribution of b is disciplined by the empirical size distribution of those contractions.
- q is disciplined by which disasters produced bill losses or inflation/default-like outcomes.
- $d = b$ is chosen to match the fact that bills performed similarly to stocks in some wartime crises.

Interpretation. The model is not estimated from equity returns. It calibrates tail-risk parameters from macroeconomic disaster history.

4.2 Why international history is important

- U.S. history alone understates disaster risk because the United States avoided massive domestic destruction in the world wars.
- Many other OECD countries experienced severe output contractions in World War I and World War II.
- Latin American and Asian countries experienced significant postwar depressions.

Interpretation. The rare-disaster probability relevant for asset pricing is not obvious from one country that had an unusually lucky twentieth century.

4.3 What the Mehra-Prescott benchmark misses

- With only normal growth shocks, the equity premium is $\theta\sigma^2$.
- With $\theta = 4$ and $\sigma = 0.02$, this is only 0.16%.
- The model also implies a real bill rate above 12%, far above observed rates.

Interpretation. The standard representative-agent model fails because normal consumption volatility is too small. Rare disasters add the missing tail risk.

4.4 What the calibration explains

- A high equity premium.
- A low expected real bill rate.
- Fat tails in long-sample growth data.
- Low expected real interest rates during wars.
- Potentially, stock return volatility if the disaster probability varies over time.

Interpretation. The paper's strength is that one mechanism speaks to several puzzles at once.

4.5 What the paper cannot fully prove

- Disaster probabilities and sizes are calibrated from limited historical samples.
- The model treats disasters as i.i.d. jumps, even though historical crises have duration and persistence.
- The baseline model has complete markets and a representative agent.
- The baseline model treats property rights on equity as secure, except in discussion, even though some historical disasters involved expropriation.
- The model does not fully estimate time-varying disaster probability p_t.
- The theory does not directly identify what investors actually believed about future disasters at each date.

Interpretation. The paper is a calibrated theory with historical discipline, not a reduced-form causal design.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Nearby tables and figures are grouped to save space and improve reading speed.

5.1 Table I and Figure I: How frequent and how large were disasters?

Read:

- Table I lists contractions of at least 15% in real per capita GDP.
- The main sample has 60 disasters after excluding 5 war aftermaths.
- The historical events are concentrated around World War I, the Great Depression, World War II, and post-WWII depressions outside the OECD.
- Figure I shows that most contractions are between 15% and 39%, but there are several very large contractions above 45%.
- The raw mean contraction is about 29%; trend-adjusted contractions average about 35%.

Economic message. The empirical distribution in Table I and Figure I is the central input to the model. The rare-disaster story is plausible only if the required disasters are not historically absurd.

TABLE I
DECLINES OF 15 PERCENT OR MORE IN REAL PER CAPITA GDP

Part A: Twenty OECD Countries in Maddison [2003]

Event	Country	Years	% fall in real per capita GDP
World War I	Austria	1913–1919	35
	Belgium	1916–1918	30
	Denmark	1914–1918	16
	Finland	1913–1918	35
	France	1916–1918	31
	Germany	1913–1919	29
	Netherlands	1913–1918	17
	Sweden	1913–1918	18
Great Depression	Australia	1928–1931	20
	Austria	1929–1933	23
	Canada	1929–1933	33
	France	1929–1932	16
	Germany	1928–1932	18
	Netherlands	1929–1934	16
	New Zealand	1929–1932	18
	United States	1929–1933	31
Spanish Civil War	Portugal	1934–1936	15
	Spain	1935–1938	31
World War II	Austria	1944–1945	58
	Belgium	1939–1943	24
	Denmark	1939–1941	24
	France	1939–1944	49
	Germany	1944–1946	64
	Greece	1939–1945	64
	Italy	1940–1945	45
	Japan	1943–1945	52
	Netherlands	1939–1945	52
	Norway	1939–1944	20
Aftermaths of wars	Canada	1917–1921	30
	Italy	1918–1921	25
	United Kingdom	1918–1921	19
	United Kingdom	1943–1947	15
	United States	1944–1947	28

Part B: Eight Latin American and Seven Asian Countries in Maddison [2003]

Event	Country	Years	% fall in real per capita GDP
World War I	Argentina	1912–1917	29
	Chile	1912–1915	16
	Chile	1917–1919	23
	Uruguay	1912–1915	30
	Venezuela	1913–1916	17

Table 1: Table I, Part A: OECD disaster contractions.

TABLE I
(CONTINUED)

Event	Country	Years	% fall in real per capita GDP
Great Depression	Argentina	1929–1932	19
	Chile	1929–1932	33
	Mexico	1926–1932	31
	Peru	1929–1932	29
	Uruguay	1930–1933	36
	Venezuela	1929–1932	24
	Malaysia	1929–1932	17
	Sri Lanka	1929–1932	15
World War II	Peru	1941–1943	18
	Venezuela	1939–1942	22
	Indonesia*	1941–1949	36
	Malaysia**	1942–1947	36
	Philippines***	1940–1946	59
	South Korea	1938–1945	59
	Sri Lanka	1943–1946	21
	Taiwan	1942–1945	51
Post-WWII Depressions	Argentina	1979–1985	17
	Argentina	1998–2002	21
	Chile	1971–1975	24
	Chile	1981–1983	18
	Peru	1981–1983	17
	Peru	1987–1992	30
	Uruguay	1981–1984	17
	Uruguay	1998–2002	20
	Venezuela	1977–1985	24
	Indonesia	1997–1999	15
	Philippines	1982–1985	18
Mean for 60 contractions (excluding 5 war aftermaths in part A)			29

Part A uses data from Maddison [2003] for twenty OECD countries for 1900–2000: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, and United States. Switzerland (−11 percent in 1915–1918 and −10 percent in 1929–1932) had no 15 percent events. With the war aftermaths excluded, the United Kingdom also lacked 15 percent events, and the largest contraction was 7 percent in 1929–1931. Satisfactory data for Ireland are unavailable until after World War II. Data for Greece are missing around World War I, 1914–1920, and also for 1901–1912.

Part B covers eight Latin American and seven Asian countries that have nearly continuous data from Maddison [2003] at least from before World War I. The sample is Argentina, Brazil, Chile, Colombia, Mexico, Peru, Uruguay, Venezuela, India, Indonesia, Malaysia, Philippines, South Korea, Sri Lanka, and Taiwan. Data for Argentina and Uruguay after 2001 are from Economist Intelligence Unit, *Country Data*. Countries that had no 15 percent events are Brazil (−13 percent in 1928–1931, −13 percent in 1980–1983), Colombia (−9 percent in 1913–1915), and India (−11 percent in 1916–1920, −12 percent in 1943–1948). Data for Peru appear to be unreliable before the mid-1920s.

Adjustments were made by Maddison to account for changes in country borders.

* No data available for 1942–1948.

** No data available for 1941–1945.

Table 2: Table I, Part B: Latin American and Asian contractions.

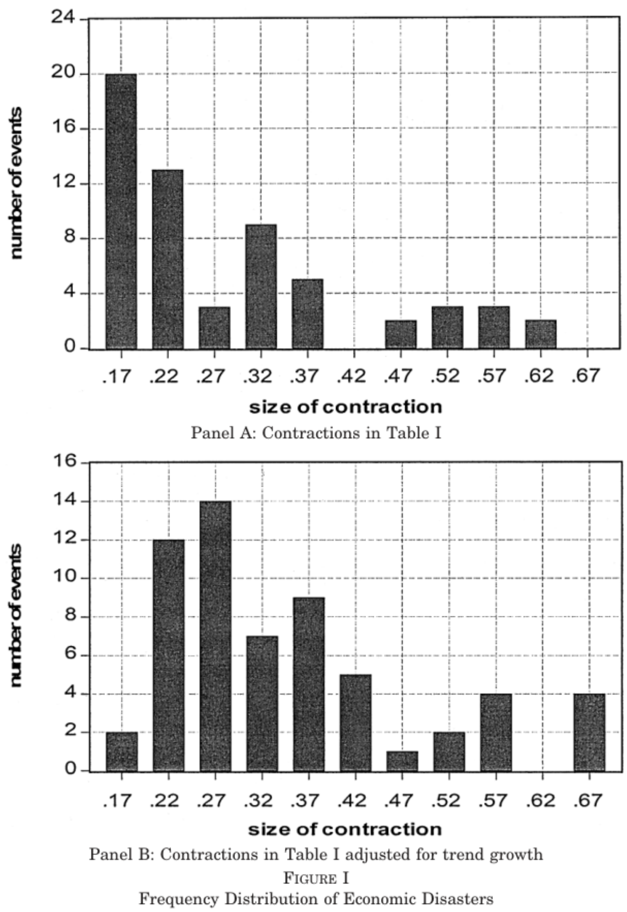

Figure 1: Figure I: frequency distribution of disaster sizes.

5.2 Table II and Table III: Crisis returns and growth-rate moments

Read:

- Table II shows that during the Great Depression, stocks performed badly and bills performed well.
- During post-WWII depressions, the same pattern mostly holds: stocks did badly, bills did comparatively well.
- During world wars, both stocks and bills often had negative real returns, making bill default/inflation risk necessary in the model.
- Table III shows that 1954-2004 G7 growth rates have kurtosis near 3, roughly consistent with normal shocks in tranquil times.
- In 1890-2004, G7 growth-rate kurtosis is much higher, especially for Germany and Japan, reflecting large disaster observations.

Economic message. The model needs two components: normal shocks for normal periods and rare disasters for fat tails. Table II also motivates why bills are partly risky in disasters.

TABLE II

STOCK AND BILL RETURNS DURING ECONOMIC CRISES

Event	Real stock return (% per year)	Real bill return (% per year)
World War I		
Austria, 1914–1918	—	−4.1
Denmark, 1914–1918	—	−6.9
France, 1914–1918	−5.7	−9.3
Germany, 1914–1918	−26.4	−15.6
Netherlands, 1914–1918	—	−5.2
Sweden, 1914–1918	−15.9*	−13.1
Great Depression		
Australia, 1928–1930	−3.6	8.2
Austria, 1929–1932	−17.3*	7.1
Canada, 1929–1932	−23.1*	7.1
Chile, 1929–1931	−22.3*	—
France, 1929–1931	−20.5	1.4
Germany, 1928–1931	−14.8	9.3
Netherlands, 1929–1933	−14.2*	5.7
New Zealand, 1929–1931	−5.6*	11.9
United States, 1929–1932	−16.5	9.3
Spanish Civil War		
Portugal, 1934–1936	13.4*	3.8
World War II		
Denmark, 1939–1945	−3.7*	−2.2
France, 1943–1945	−29.3	−22.1
Italy, 1943–1945	−33.9	−52.6
Japan, 1939–1945	−2.3	−8.7
Norway, 1939–1945	1.7*	−4.5
Post-WWII Depressions		
Argentina, 1998–2001	−3.6	9.0
Chile, 1981–1982	−37.0*	14.0
Indonesia, 1997–1998	−44.5	9.6
Philippines, 1982–1984	−24.3	−5.0
Thailand, 1996–1997**	−48.9	6.0
Venezuela, 1976–1984	−8.6*	—

Real rates of return on stocks and bills are for periods corresponding to the economic contractions in Table I. For the Great Depression and post-World War II depressions, the periods are from the start of the depression to the year before the rebound in the economy. For the world wars, the periods are 1914–1918 and 1939–1945, except for France and Italy where data for 1939–1942 are problematic. The returns are averages of arithmetic annual real rates of return based on total-return indexes, except the entries marked by an asterisk used stock-price indexes. Real values were computed by using consumer price indexes. Data are from Global Financial Data, except that the nominal return on bills for Indonesia in 1997–1998 is for money-market rates from EIU *Country Data*.

* Based on stock-price indexes, rather than total-return indexes.

** Thailand's contraction of real per capita GDP by 14 percent for 1996–1998 falls just short of the 15 percent criterion used in Table I.

Table 3: Table II: stock and bill returns during crises.

TABLE III

GROWTH RATES OF REAL PER CAPITA GDP IN G7 COUNTRIES

	Canada	France	Germany*	Italy	Japan	U. K.	U. S.
	Growth rate of real per capita GDP, 1890–2004						
Mean	0.021	0.020	0.019	0.022	0.027	0.015	0.021
Standard deviation	0.051	0.069	0.090	0.059	0.082	0.030	0.045
Kurtosis	5.4	5.4	40.6	10.4	49.0	5.8	5.8
	Growth rate of real per capita GDP, 1954–2004						
Mean	0.022	0.026	0.027	0.030	0.043	0.021	0.021
Standard deviation	0.023	0.017	0.024	0.022	0.034	0.018	0.022
Kurtosis	3.4	2.5	3.9	2.8	2.4	3.1	2.6

* Data missing for Germany in 1918–1919.

Except for the United States, data are from Maddison [2003], updated through 2004 using information from Economist Intelligence Unit, *Country Data*. The U. S. sources are noted in the text.

Table 4: Table III: G7 growth moments.

5.3 Table IV and Table V: Empirical target and model calibration

Read:

- Table IV shows the empirical equity premium target: about 7% in both long samples and 1954-2004.
- Long-sample real stock returns average 7.1%; real bill returns average about -0.1%; the spread is 7.2%.
- In 1954-2004, real stock returns average 8.7%, bill returns 1.7%, and the spread 7.0%.
- Table V column (1) shows the no-disaster failure: equity premium 0.16% and bill rate 12.7%.
- Table V column (2) baseline with disasters gives unlevered premium 3.6%, bill rate 3.5%, P/E ratio 19.6, and levered premium 5.4%.
- With $\theta = 3$, the levered premium falls to 2.4%, showing that the results are sensitive to risk aversion.
- Raising p from 0.017 to 0.025 raises the levered premium from 5.4% to 7.8% and lowers the bill rate to -0.7%.
- Lowering q from 0.4 to 0.3 raises the levered premium to 6.3% because bills become better disaster hedges.

Economic message. Table IV states the asset-pricing target. Table V shows that rare disasters move the Lucas-tree model from a spectacular failure to the right quantitative neighborhood.

TABLE IV

STOCK AND BILL RETURNS FOR G7 COUNTRIES
(AVERAGES OF ARITHMETIC ANNUAL RETURNS, STANDARD DEVIATIONS IN PARENTHESES)

Country & time period	Real stock return	Real bill return	Spread
1. Long samples			
Canada, 1934–2004	0.074 (0.160)	0.010 (0.036)	0.063 (0.163)
France, 1896–2004	0.070 (0.277)	−0.018 (0.095)	0.088 (0.279)
Germany, 1880–2004*	0.052 (0.306)	−0.009 (0.142)	0.061 (0.269)
Italy, 1925–2004	0.063 (0.296)	−0.009 (0.128)	0.072 (0.283)
Japan, 1923–2004	0.092 (0.296)	−0.012 (0.138)	0.104 (0.271)
U. K., 1880–2004	0.063 (0.183)	0.016 (0.055)	0.047 (0.179)
U. S., 1880–2004	0.081 (0.190)	0.015 (0.047)	0.066 (0.191)
Means	0.071 (0.240)	−0.001 (0.092)	0.072 (0.234)
2. 1954–2004			
Canada	0.074 (0.165)	0.024 (0.024)	0.050 (0.168)
France	0.091 (0.254)	0.019 (0.029)	0.072 (0.252)
Germany	0.098 (0.261)	0.018 (0.015)	0.080 (0.260)
Italy	0.067 (0.283)	0.016 (0.034)	0.051 (0.279)
Japan	0.095 (0.262)	0.012 (0.037)	0.083 (0.253)
U. K.	0.097 (0.242)	0.018 (0.033)	0.079 (0.250)
U. S.	0.089 (0.180)	0.014 (0.021)	0.076 (0.175)
Means	0.087 (0.235)	0.017 (0.028)	0.070 (0.234)

* Data missing for Germany in 1923–1924 and 1945.

Indexes of cumulated total nominal returns on stocks and government bills or analogous paper are from Global Financial Data. See Taylor [2005] for a discussion. Nominal values for December of each year are converted to real values by dividing by consumer price indexes. Annual real returns are computed arithmetically based on December-to-December real values. CPI data since 1970 are available online from Bureau of Labor Statistics and OECD. Earlier data are from Bureau of Labor Statistics, U. S. Department of Commerce [1975], Mitchell [1980, 1982, 1983], and Mitchell and Deane [1962]. The German data for the CPI have breaks corresponding to the hyperinflation in 1923–1924 and the separation into East and West in 1945. Therefore, real returns are unavailable for these years. German data on dividends are missing for 1942–1952. Hence

Table 5: Table IV: G7 stock and bill returns.

TABLE V
CALIBRATED MODEL FOR RATES OF RETURN

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Parameters						
	No disasters	Baseline	Low θ	High p	Low q	Low γ	Low ρ
θ (coeff. of relative risk aversion)	4	**4**	**3**	4	4	4	4
σ (s.d. of growth rate, no disasters)	0.02	**0.02**	0.02	0.02	0.02	0.02	0.02
ρ (rate of time preference)	0.03	**0.03**	0.03	0.03	0.03	0.03	**0.02**
γ (growth rate, deterministic part)	0.025	**0.025**	0.025	0.025	0.025	**0.020**	0.025
p (disaster probability)	**0**	**0.017**	0.017	**0.025**	0.017	0.017	0.017
q (bill default probability in disaster)	**0**	**0.4**	0.4	0.4	**0.3**	0.4	0.4
	Variables						
Expected equity rate	0.128	0.071	0.076	0.044	0.071	0.051	0.061
Expected bill rate	0.127	0.035	0.061	−0.007	0.029	0.015	0.025
Equity premium	0.0016	0.036	0.016	0.052	0.042	0.036	0.036
Expected equity rate, conditional	0.128	0.076	0.081	0.052	0.076	0.056	0.066
Face bill rate	0.127	0.037	0.063	−0.004	0.031	0.017	0.027
Equity premium, conditional	0.0016	0.039	0.019	0.056	0.045	0.039	0.039
Price-earnings ratio	9.7	19.6	17.8	37.0	19.6	27.8	24.4
Expected growth rate	0.025	0.020	0.020	0.018	0.020	0.015	0.020
Expected growth rate, conditional	0.025	0.025	0.025	0.025	0.025	0.020	0.025
	Levered results (debt-equity ratio is $\lambda = 0.5$)						
Expected equity rate	0.129	0.089	0.084	0.071	0.092	0.069	0.079
Equity premium	0.0024	0.054	0.024	0.078	0.063	0.059	0.054
Expected equity rate, conditional	0.129	0.096	0.091	0.080	0.099	0.076	0.086
Equity premium, conditional	0.0024	0.059	0.028	0.084	0.068	0.059	0.059

Cells show the calibrated model's rates of return and growth rates, based on the indicated parameter values. The distribution of disaster sizes b comes from Table I (Figure I, panel A). The expected rate of return on equity is in (9). The expected rate of return on bills is in (12). The equity premium is the difference between these two rates. The expected rate of return on equity conditioned on no disasters is in (10). The face bill rate is in (11). The equity premium conditioned on no disasters is the difference between these two rates. The price-earnings ratio is in (17). The expected growth rate is in (18), and the expected growth rate conditioned on no disasters is in (19). The levered expected rate of return on equity is in (22), and the levered equity premium is in (23). Conditioning on no disasters raises the levered expected return on equity by $p \cdot Eb \cdot (1 + \lambda - q\lambda)$ and the levered equity premium by $(1 + \lambda) \cdot p \cdot (1 - q) \cdot Eb$.

Table 6: Table V: calibrated model results.

5.4 Figure II and Table VI: How much risk aversion, disaster probability, and duration are needed?

Read:

- Figure II plots combinations of p and θ needed to generate a levered equity premium of 7%.
- With baseline $p = 0.017$, the historical disaster-size distribution requires $\theta \approx 4.3$.
- If all disasters were $b = 0.5$, the required θ falls to about 3.3.
- If all disasters were $b = 0.25$, the required θ rises to about 10.
- With $\theta = 4$, the historical distribution requires $p \approx 0.022$.
- Table VI shows that if trend-adjusted b is used, the unlevered premium rises, but finite disaster duration lowers it.
- For $\theta = 4$, trend-adjusted b with $T = 0$ gives a 6.4% unlevered premium; with $T = 3$, it gives 4.6%; with $T = 5$, it gives 3.8%.

Economic message. The model does not need infinitely catastrophic disasters if risk aversion is around 4 and the historical distribution of disaster sizes is used. Duration matters, but it does not overturn the mechanism.

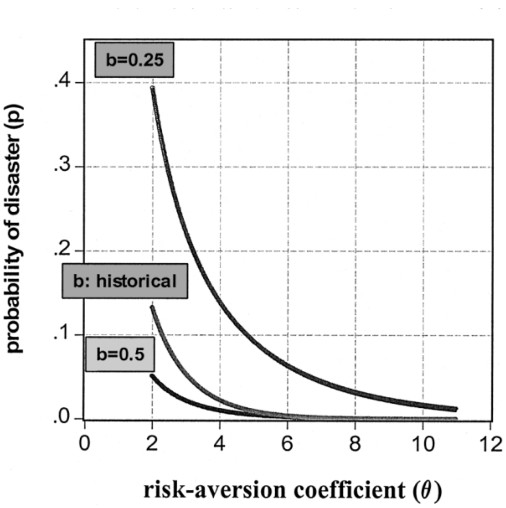

FIGURE II
Isopremium Curves

The graphs show the combinations of the disaster probability (p) and the coefficient of relative risk aversion (θ) needed to generate a levered equity premium of 0.07 in (23) with a leverage coefficient, λ, of 0.5. The middle graph uses the historical frequency of disaster sizes b, given in Figure I, panel A. In the lowest graph, b is constant at 0.5, and in the upper graph, b is constant at 0.25.

the baseline values from Table V, column (2)—in particular, the debt-equity ratio is $\lambda = 0.5$, the contingent probability of default

Figure 2: Figure II: isopremium curves.

TABLE VI
EFFECT OF PERIOD LENGTH ON EQUITY AND BILL RATES

(1) b detrended?	(2) Period length T (years)	(3) Expected equity rate (unlevered)	(4) Expected bill rate	(5) Equity premium (unlevered)
		$\theta = 4$		
No	0	0.071	0.035	0.036
Yes	0	0.044	−0.020	0.064
Yes	3	0.054	0.008	0.046
Yes	5	0.059	0.020	0.038
		$\theta = 3$		
No	0	0.076	0.061	0.016
Yes	0	0.066	0.041	0.025
Yes	3	0.068	0.047	0.021
Yes	5	0.070	0.051	0.019

The cells show the expected equity and bill rates and the equity premium in the discrete-period model with period length T. Column (1) indicates whether the distribution of disaster sizes b is computed from raw data (Figure I, panel A) or trend-adjusted values (Figure I, panel B). The expected equity rate in column (3) comes from (25). The expected bill rate in column (4) comes from (26). The equity premium in column (5) is the difference between these two rates.

Table 7: Table VI: period length effects.

5.5 Figure III and Table VII: Expected real interest rates in wars and depressions

Read:

- Figure III shows U.S. expected real short-term rates from 1859 to 2004.
- Expected real rates tend to be low during the Civil War, World War I, World War II, and the early Korean War.
- Table VII shows strongly negative expected real returns in World War I in 1917 and 1918, and in World War II in 1942-1945.
- During the Great Depression, measured expected real rates are high in 1931-1933 because expected deflation is estimated to be negative; Barro argues this likely overstates rationally expected deflation.
- From 1934 onward, nominal rates near zero and positive expected inflation generate low expected real rates.

Economic message. A rise in disaster probability p lowers the bill rate. This provides a rare-disaster explanation for the wartime real-rate puzzle: wars are periods when future disaster risk becomes more salient.

FIGURE III

Expected Real Interest Rate on U. S. T-Bills/Commercial Paper, 1859–2004

Data on nominal returns on U. S. Treasury Bills (1922–2004) and Commercial Paper (1859–1921) are from Global Financial Data. See the notes to Table IV. From 1947–2004, expected real returns are nominal returns less the Livingston expected inflation rate for the CPI (using six-month-ahead forecasts from June and December). For 1859–1946 the expected real return is the nominal return less a constructed estimate of expected inflation derived from a first-order autoregression of CPI inflation rates for 1859–1946. The CPI data are from Bureau of Labor Statistics (January values since 1913, annual averages before 1913) and U. S. Department of Commerce [1975].

Figure 3: Figure III: U.S. expected real interest rates.

TABLE VII
INTEREST AND INFLATION RATES DURING WARS AND THE GREAT DEPRESSION IN THE UNITED STATES

Year	Nominal return	Expected inflation rate	Expected real return
Civil War			
1860	0.070	0.006	0.063
1861 (start of war)	0.066	0.026	0.039
1862	0.058	0.063	−0.005
1863	0.051	0.082	−0.031
1864	0.062	0.128	−0.066
1865 (end of war)	0.079	0.050	0.029
Spanish-American War			
1897	0.018	0.015	0.004
1898 (year of war)	0.021	0.006	0.015
World War I			
1914 (start of war)	0.047	0.021	0.026
1915	0.033	0.011	0.022
1916	0.033	0.026	0.007
1917 (U. S. entrance)	0.048	0.075	−0.028
1918 (end of war)	0.059	0.116	−0.057
Great Depression			
1929	0.045	0.000	0.044
1930 (start of depression)	0.023	0.006	0.016
1931	0.012	−0.038	0.050
1932	0.009	−0.059	0.068
1933 (worst of depression)	0.005	−0.057	0.062
1934	0.003	0.022	−0.020
1935	0.002	0.025	−0.023
1936	0.002	0.015	−0.014
1937 (onset of sharp recession)	0.003	0.018	−0.016
1938	0.001	0.012	−0.012
World War II			
1939 (start of war)	0.000	−0.005	0.006
1940	0.000	0.005	−0.005
1941 (U. S. entrance)	0.001	0.014	−0.012
1942	0.003	0.072	−0.068
1943	0.004	0.053	−0.049
1944	0.004	0.024	−0.021
1945 (end of war)	0.004	0.021	−0.017
Korean War			
1950 (start of war)	0.012	0.014	−0.002
1951	0.016	0.026	−0.010
1952	0.017	0.005	0.012
1953 (end of war)	0.019	−0.009	0.028
Vietnam War			
1964	0.036	0.011	0.025
1965 (start of main war)	0.041	0.012	0.029
1966	0.049	0.018	0.031
1967	0.044	0.022	0.022
1968	0.055	0.029	0.026
1969	0.069	0.032	0.037
1970	0.065	0.036	0.029
1971	0.044	0.035	0.008
1972 (end of main war)	0.042	0.033	0.009
Gulf War			
1990	0.077	0.039	0.038
1991 (year of war)	0.054	0.035	0.020
1992	0.035	0.034	0.001
Afghanistan-Iraq War			
2000	0.058	0.025	0.033
2001 (September 11)	0.033	0.025	0.008
2002 (start of Afghanistan war)	0.016	0.022	−0.006
2003 (start of Iraq war)	0.010	0.017	−0.006
2004	0.014	0.018	−0.004

TABLE VII
(CONTINUED)

Nominal returns on U. S. Treasury Bills or commercial paper (before 1922) are calculated as in Table IV. The expected inflation rate for the CPI is constructed as described in the notes to Figure III. The expected real return is the difference between the nominal return and the expected inflation rate.

6 Short summary

- Barro revives and calibrates the Rietz rare-disaster explanation for the equity premium.
- The model keeps a representative agent, isoelastic utility, complete markets, and a Lucas-tree economy.
- Output growth has normal shocks plus rare downward jumps.
- The paper calibrates rare disasters from twentieth-century international data rather than choosing disaster sizes arbitrarily.
- The baseline disaster probability is about 1.7% per year.
- Disaster contractions range from 15% to 64% of real per capita GDP.
- The mean raw contraction is about 29%, and the trend-adjusted mean is about 35%.
- Bills partly default in disasters with conditional probability $q = 0.4$, mostly representing inflationary erosion during wars.
- Without rare disasters, the model produces a trivial equity premium and unrealistically high bill rate.
- With rare disasters and $\theta = 4$, the model produces a 3.6% unlevered premium and a 5.4% levered premium.
- Increasing the effective leverage coefficient or disaster probability can move the premium closer to the empirical 7%.
- The model also explains why real bill rates can be low when disaster probability is high.
- U.S. wartime expected real rates provide suggestive support for the bill-rate prediction.
- Persistent variation in disaster probability can potentially explain volatile stock prices.
- The capital-formation extension shows that the mechanism does not depend on a fixed-tree economy.

7 Conclusion

7.1 Core conclusion

Rare disasters can rationalize major asset-pricing facts that are difficult to explain with normal consumption risk alone. Historically observed disasters are large enough and frequent enough to make this explanation plausible.

7.2 Economic intuition

Investors care most about payoffs when consumption is very low. Stocks perform poorly in those states, while bills often perform better unless they default through inflation or other mechanisms. This makes equity require a high premium and makes safe assets carry low expected returns.

7.3 Takeaway

The paper's lasting message is that asset prices are shaped by events that may not appear in a short sample. Tail risk is not a small correction to standard asset-pricing models; it can be the main force behind the equity premium, the risk-free rate, and real interest rates during crises.